WHITEPAPER / PITCH · BOT CHAIN ECOSYSTEM SUPPORT PROGRAM

AgentRail

A Simulate → Policy → Execute → Verify control plane for AI agents on BOT Chain.

Version 1.0 · 16 August 2026 · Public testnet live

Applicant: Ujjwal Dubey · ujjwal07dubey@gmail.com · github.com/ujjwaldubey1/AgentRail

Abstract

BOT Chain is an AI-native EVM L1. Language models can already propose payments. They must not hold an unrestricted treasury key. Prompt injection, hallucinated calldata, or a runaway loop will drain a normal EOA.

AgentRail is a BOT-native control plane for that gap. Funds sit in a vault. Two keys, never the same address: an owner who sets policy, and an agent who may only call `proposeAndExecute`. Clients `eth_call` `simulate(intent)` before they sign. If policy fails, execution does *not* revert. It emits `Decision(allowed=false, reason)` so blocked attacks are permanent on-chain receipts. After inclusion, the client verifies logs and balances. A transaction hash is not payment.

The product is live on BOT Chain Testnet (chain ID 968). We are not launching a token. We ask for milestone-based support so other BOT applications can run agents without handing them the treasury.

1. Pitch

Agents should not execute blindly. AgentRail is the rail they execute on.

Category	AI agent infrastructure / permissioned payments
Chain	BOT Chain Testnet, ID 968
Native asset	BOT (vault + gas)
Status	Live contracts + dApp + on-chain proofs
Ask	Milestone grant + BOT Labs matching (paymaster, explorer, listing)

2. Problem

BOT Labs' agent-payment checklist: typed intent; simulate before send; deterministic policy outside the model; execute only if policy passes; verify receipt/events/balances; record refusals. The L1 ships EVM, RPC, explorer, and an EOA paymaster spec. It does not yet ship this control plane.

If BOT sits in an agent EOA, a prompt-injected recipient, a 1000 BOT hallucination, or a hidden revert can drain or obscure the truth. That is a hot wallet, not an agent architecture.

3. Solution

3.1 Split keys

Role	Holds	Can do
Owner	Treasury policy	Deposit, withdraw, pause, caps, allowlists, rotate agent
Agent	Gas only	`proposeAndExecute(Intent)`
Visitor	Own wallet	Read vault and proofs; cannot spend another vault

Owner and agent cannot be the same address. Compromising the agent cannot empty the vault past policy.

3.2 Loop

Read → Simulate → Execute → Verify. Typed `Intent { to, token, amount, data, deadline, actionId }`. Simulate returns `OK`, `CAP`, `TARGET`, and related codes. Failed policy emits `Decision` and returns false; only a non-agent caller reverts `Unauthorized()`.

4. Policy (v1)

Paused, expiry, deadline, replay lock, token allowlist, destination allowlist, optional selector allowlist, per-tx cap, UTC-day cap. Live demo: 0.5 BOT per tx, 2.0 BOT per UTC day, native BOT, one payee.

5. New users

The public dApp is one proof vault. A stranger who connects is role Other: they can see stats and proofs, not spend this vault. A real user deploys (or will deploy via factory) their own vault, deposits as owner, and gives the agent address — not the owner key — to a bot. There is no AgentRail backend. The contract is the backend.

6. Live proof (chain 968)

Vault	0x254AceA1E7411EA396a6a8802316206cFfB14171
Payee	0x8bf5319Db9cD308D52bA8f4a6c04267FfaA08049
Owner	0x184E46634F2E21d88365ffC2bF58a83e315f3c8c
Agent	0xb0Bb213DC381287c6A0D0A279ac9Cf423e7A340e
RPC	https://rpc.bohr.life
Repo	github.com/ujjwaldubey1/AgentRail

Demo	Simulate	Tx
Allow 0.1 BOT	OK	0xbd15aefe…
Cap drain 1000 BOT	CAP	0x377f1766…
Inject recipient	TARGET	0x4ade5df4…

UI replay (16 Aug 2026): deposit 0x5ce5e694, allow 0x0d8289a8, cap 0x0a75e727, inject 0x435ca100. Explorer Success on blocked txs is correct: the refusal is included, the payment is not.

Tests: `forge test` — 13 passing.

7. Why BOT Chain

Native tBOT in the vault; BOT gas on every agent action; official RPC/explorer. Next: EOA paymaster, B DEX as an allowlisted router, CaryPact receipts. Differentiation: pre-flight simulate, no-revert blocked receipts, verify after inclusion — not another DEX or wallet.

8. 90-day milestones

Days	Deliverable	Metric
0–30	SDK, create-vault / policy UI, documented reasons	≥50 Decision events; ≥10 unique blocked attacks
31–60	Paymaster; session-key rotation; B DEX swap demo	≥200 agent txs
61–90	Mainnet after review; identity stub; grant report	Mainnet vault + BOT gas attributable to AgentRail

9. Ask

Milestone grant (engineering, audit, gas). BOT Labs account manager for paymaster, explorer, listing. Ecosystem certification when mainnet is ready. Not a token launch. Success is other teams settling agent payments so BOT fee flow rises without new inflation.

10. Risks

v1 is allowlist + caps. Public dApp is one demo vault until factory. Testnet only. Agent still needs gas until paymaster. The rail refuses bad intents; it does not make the model honest.

11. Contact

Ujjwal Dubey · ujjwal07dubey@gmail.com · x.com/ujjwal07dubey · github.com/ujjwaldubey1/AgentRail

Primary wallet (owner EOA, not the contract): `0x184E46634F2E21d88365ffC2bF58a83e315f3c8c`

Network: BOT Chain Testnet · RPC https://rpc.bohr.life · Chain ID 968 · Symbol BOT · Explorer https://scan.bohr.life. Do not use ChainList 968 (Datagram).